

2e. Design Engineering (H404/01 and H404/02)

The subject content of this component is focused towards electronics and engineered products and systems and their analysis in respect of:

- materials and components, and their selection and uses in products/systems
- wider issues affecting design decisions.

It is essential that materials, components and systems are studied from the perspective of analysing modern engineered products. Learners should gain practical experience of using materials, components and

systems and, where possible, the content which follows should be learned through applied practical activities, set within realistic design scenarios.

The aim of the component is to give learners a framework for analysing existing products/systems that enables them to make considered selections of appropriate materials, components, systems and manufacturing processes when designing.

The component brings together the knowledge, understanding and skills acquired in the NEA.

1. Identifying requirements		
Considerations		Maths & Science
PRINCIPLES	1.1 What can be learnt by exploring contexts that design solutions are intended for?	
	a. Understand that all design practice is context dependent and that investigations are required to identify what makes a context distinct in relation to: i. environment and surroundings ii. user requirements iii. economic and market considerations iv. product opportunities.	